

HAIKANG DENG

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EDUCATION

- University of California, Los Angeles** Sep 2024 - Present
Ph.D. in Computer Science
Advisor: Nanyun Peng
- University of North Carolina, Chapel Hill** Aug 2019 - May 2023
B.S. in Computer Science & Statistics GPA: 3.96/4.0
Advisor: Colin Raffel, Guanting Chen

PUBLICATIONS

- [1] Reward-Augmented Decoding: Efficient Controlled Text Generation With a Unidirectional Reward Model
Haikang Deng and Colin Raffel
Empirical Methods in Natural Language Processing (EMNLP), 2023
- [2] Large Language Models Struggle to Learn Long-Tail Knowledge
Nikhil Kandpal, **Haikang Deng**, Adam Roberts, Eric Wallace, and Colin Raffel
International Conference on Machine Learning (ICML), 2023

RESEARCH & PROFESSIONAL EXPERIENCE

- University of North Carolina, Chapel Hill** Aug 2022 - Mar 2024
Research Assistant Advised by Prof. Colin Raffel
- Conducted research on human feedback alignment, controlled text generation, and retrieval augmentation to foster efficient methods on preference optimization
 - Trained expert model routing and tagging in a contrastive manner with signals from a reward model
- Amazon** Dec 2023 - Jul 2024
Software Dev Engineer Bellevue, WA
- Composed strategy to trigger pre-compute of products based on given marketplace, merchant, and store utilization score
 - Wrote a tool page for PMs to update up to 1M products' lead time in one go
 - Built the Horizonte service for Local Landing Page which displays local products available for pick up
- Lenovo** May 2021 - Aug 2021
Software Engineer Intern Beijing, China
- Trained Encoder-Decoder LSTM for anomaly detection on time series data
 - Participated in the design of Control Chart and Anomaly Detection Module
 - Performed model tuning and data grouping which improved f1 score from 0.41 to 0.48

PROJECTS

- Performance Degradation Under Format Constraints** Nov 2024 - Present
- Explore LLM performance degradation under format constraints for structured and unstructured generation
 - Delegate format control to auxiliary module to mitigate performance drop
- Neurally Decomposed Oracle for Preference Optimization** Jul 2024 - Present
- Derive training objective from NADO and adapt to preference data
 - Analyze model-generated response as compared to SOTA methods such as DPO
 - Interpolate and extrapolate policy model to explore the potential of expedited training
- Embedding-Based Routing and Tagging of PEFT Modules** Oct 2023 - Mar 2024
- Trained a router that routes a query to a PEFT module using input embeddings
 - Examined performance gap between held-in and held-out data routing
 - Enabled zero-shot PEFT module addition to the pool of modules
- Tracing Model-Generated Wrong Answers** Nov 2022 - Feb 2023
- Explored correlation between QA accuracy and the number of relevant documents that contain negative co-occurrences
 - Discovered that incorrect generation is not a result of more association of wrong answer and query entities

ACADEMIC ACHIEVEMENTS

- Successful Participant, MCM 2021 Feb 2021
9th Place in China/27th Place Worldwide in AAPT Physics Bowl Apr 2018